

Side-by-side comparison: AURORA-ADAS vs Tesla Autopilot

Dimension	AURORA-ADAS v2.0 (Aerial)	Tesla Autopilot (Ground, adapted)	Waymo Driver (Ground, adapted)
Platform	Multirotor / aerial robot	Passenger car	Robotaxi / AV
Primary Sensors	Cameras, IMU, GPS, optional depth	Cameras, radar, ultrasonics	LiDAR, cameras, radar
Perception Backbone	CNN (DroNet) + ViT + multimodal transformer	CNN + vision networks	LiDAR-centric 3D detection + vision
World Model	ViT latent dynamics + predictive free-space	Temporal vision models + occupancy networks	HD maps + dynamic object tracking
Localization	VIO + EKF/UKF + factor-graph SLAM	Visual odometry + GPS + map matching	Multi-sensor localization + HD maps
Mapping	Local occupancy + global voxel + semantic layers	Occupancy networks + lane/road models	HD semantic maps + dynamic layers
Planning	Global planner + risk-sensitive MPC + Bayesian RL	Lane-level planning + trajectory optimization	Hierarchical planning (route, behavior, motion)
Uncertainty Handling	Explicit: Bayesian RL, risk-sensitive MPC, probabilistic safety	Implicit in networks and heuristics; limited explicit probabilistic safety	Strong emphasis on uncertainty in perception and prediction, but proprietary
Safety Layer	Probabilistic constraints, confidence checks, geofencing, emergency hover/land	Safety supervisor, driver monitoring, fallback braking	Redundant safety systems, conservative behavior, multi-layer validation
Sim-to-Real Strategy	Domain randomization + synthetic aerial environments (AirSim, CARLA)	Internal simulation + fleet data	Massive simulation + fleet data, HD map validation
Operational Domain	3D aerial space (indoor/outdoor, altitude bands)	2D road network, lanes, traffic rules	2D road network with rich semantics and rules
Autonomy Style	Hybrid: learned priors + model-based planning in 3D	Strongly learned perception + model-based planning in 2D	Heavy model-based planning + rich perception and prediction